



2016 Assessment Task 4 (Trial Examination)

Thursday, 4 August 2016

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

- Mark your answers on the answer grid provided

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

BOSTES NUMBER: **# BOOKLETS USED:**

[illegible]

Section I**10 marks****Attempt Questions 1 - 10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1-10

1 What is the value of $\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^9$?

- (A) -1
- (B) 0
- (C) 1
- (D) 2

2 A particle in a straight line so that its velocity at any particular time is given by $v = k(a - x)$, where x is its displacement from a given point O.

The particle is initially at O.

Which of the following gives an expression for x ?

- (A) $x = a(1 - e^{kt})$
- (B) $x = a(1 + e^{kt})$
- (C) $x = a(1 - e^{-kt})$
- (D) $x = a(1 + e^{-kt})$

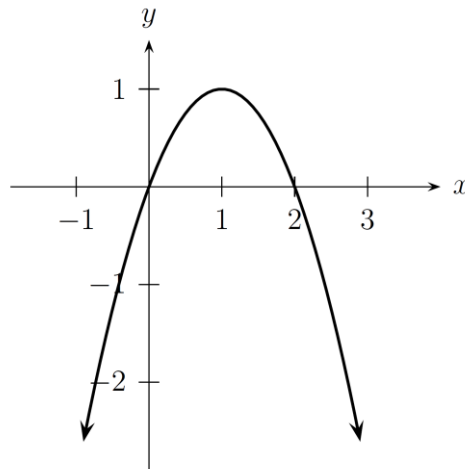
3 Let $z = 2 - 3i$. What is the value of z^{-1} ?

- (A) $-\frac{1}{5}(2 + 3i)$
- (B) $\frac{1}{13}(2 + 3i)$
- (C) $\frac{1}{5}(2 - 3i)$
- (D) $\frac{1}{13}(2 - 3i)$

4 The eccentricity of the ellipse $3x^2 + 5y^2 - 15 = 0$

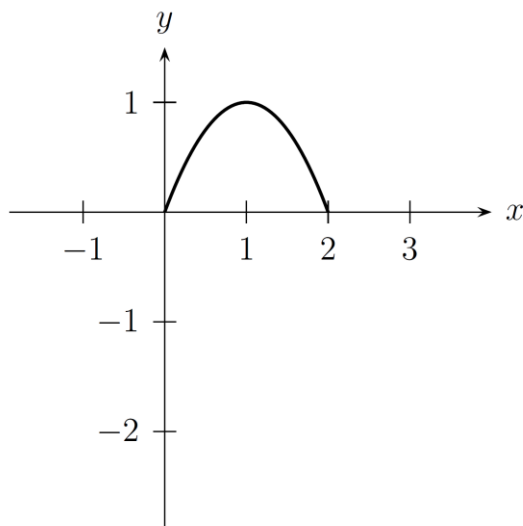
- | | |
|--------------------------|--------------------------|
| (A) $\sqrt{\frac{5}{2}}$ | (B) $\sqrt{\frac{2}{5}}$ |
| (C) $\sqrt{\frac{8}{5}}$ | (D) $\sqrt{\frac{5}{8}}$ |

- 5 The diagram below shows the graph of the function $y = f(x)$.

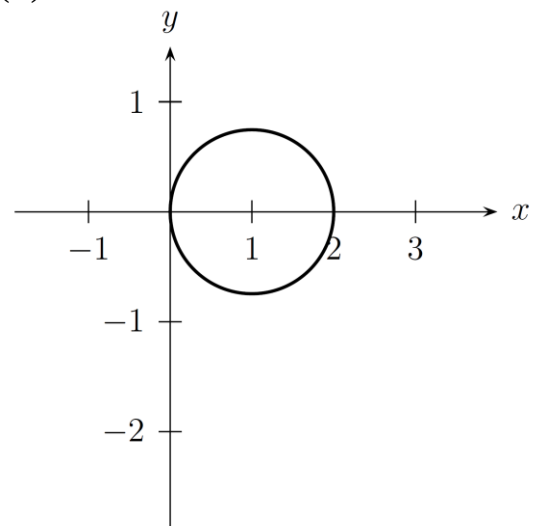


Which of the following is the graph of $y = \sqrt{f(x)}$?

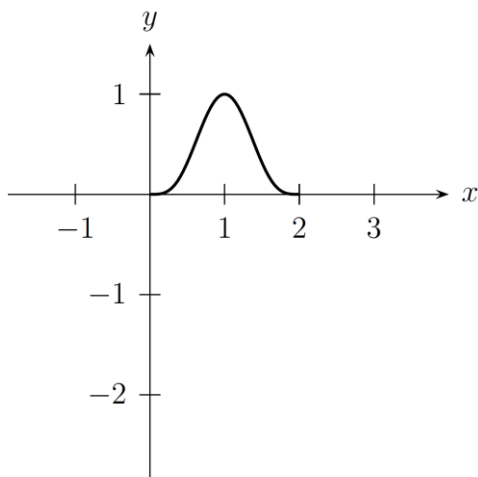
(A)



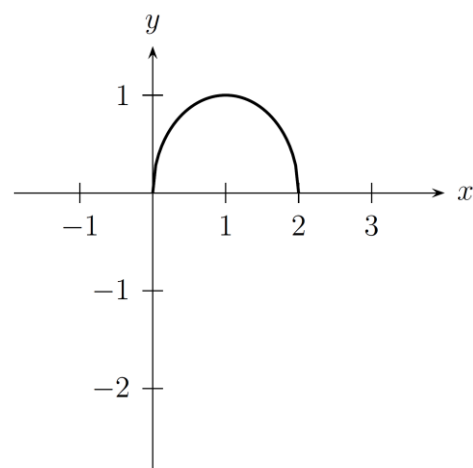
(B)



(C)

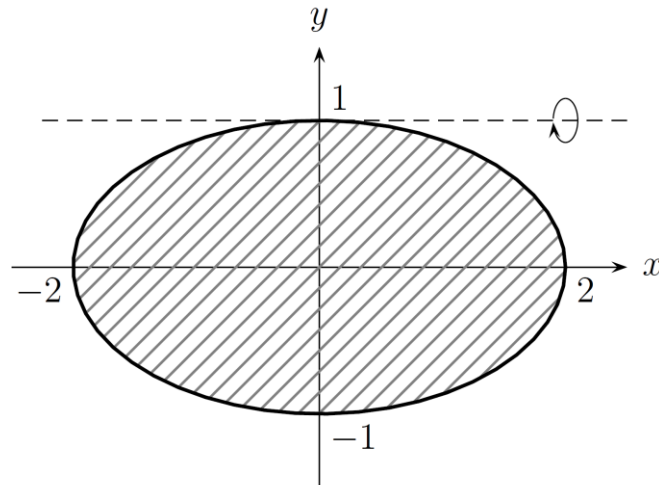


(D)



- 6 What is the value of $\int_0^2 \sqrt{\frac{x}{4-x}} dx$ using the substitution $x = 4\sin^2 \theta$?
- (A) 0.75π
 (B) $\pi - 2$
 (C) $\pi + 6$
 (D) $3\pi - 8$
- 7 What are the values of real numbers p and q such that $1-i$ is a root of the equation $z^3 + pz + q = 0$?
- (A) $p = -2$ and $q = -4$
 (B) $p = -2$ and $q = 4$
 (C) $p = 2$ and $q = -4$
 (D) $p = 2$ and $q = 4$
- 8 The points $P(cp, \frac{c}{p})$ and $Q(cq, \frac{c}{q})$ lie on the same branch of the hyperbola $xy = c^2$ ($p \neq q$)
 The tangents at P and Q meet at the point T . What is the equation of the normal to the hyperbola at P ?
- (A) $p^2x - py + c - cp^4 = 0$
 (B) $p^3x - py + c - cp^4 = 0$
 (C) $x + p^2y - 2c = 0$
 (D) $x + p^2y - 2cp = 0$
- 9 A car of mass m is travelling at a constant velocity V_0 m/s. The brakes are applied, resulting in a constant deceleration until the car comes to a stop in l m. What is the force exerted by the brakes?
- (A) $\frac{mV_0^2}{2l}$
 (B) $\frac{mV_0^2}{l}$
 (C) $-\frac{mV_0^2}{2l}$
 (D) $-\frac{mV_0^2}{l}$

10 The diagram below shows the ellipse



The ellipse is rotated around the line $y = 1$ to form a solid.

Which integral represents the volume of the torus?

(A) $4\pi \int_{-1}^1 (1-y)\sqrt{1-y^2} dy$

(B) $8\pi \int_{-1}^1 (1-y)\sqrt{1-y^2} dy$

(C) $4\pi \int_{-1}^1 y\sqrt{1-y^2} dy$

(D) $8\pi \int_{-1}^1 y\sqrt{1-y^2} dy$

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	Marks
(a) (i) Show that $(1-2i)^2 = -3-4i$.	1
(ii) Hence solve the equation $z^2 - 5z + (7+i) = 0$.	2
(b) Let $z = 2 + 2i$	
(i) What is the exact value of $ z $ and $\arg z$?	2
(ii) Find z^3 in $x + iy$ form by using De Moivre's theorem	2
(c) Let two complex numbers be $z_1 = 2(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12})$ and $z_2 = 2i$.	
(i) On an Argand diagram sketch the vectors OA and OB to represent z_1 and z_2 respectively.	1
(ii) Draw the vectors $z_1 + z_2$ and $z_1 - z_2$ on the same Argand diagram.	1
(iii) What are the exact values of $\arg(z_1 + z_2)$ and $\arg(z_1 - z_2)$?	2
(d) (i) Find the three cubic roots of -1 by solving the equation $z^3 + 1 = 0$.	2
(ii) Let α be a cubic root of -1 , where α is not real. Show that $\alpha^2 = \alpha - 1$.	1
(iii) Hence simplify $(1-\alpha)^6$.	1

Question 12 (15 marks)**Marks**

- (a) If α, β, γ are the roots of the polynomial equation $3x^3 - 5x^2 - 4x + 3 = 0$, **1**
Find the equation with roots $\alpha - 1, \beta - 1, \gamma - 1$.

- (b) The polynomial $P(x) = x^4 + ax^2 + bx + 28$ has a double root at $x = 2$. **2**
What are the values of a and b ?

- (c) The point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b > 0$.

PQ is a diameter of the ellipse. The tangent to the ellipse at P meets the vertical through Q at R .

- (i) Prove the equation of the tangent is $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$. **2**
(ii) Show the ratio of the area of $\triangle PQR$ to the area of the ellipse is **3**
 $2: \pi |\tan \theta|$.

You may use the fact that the area of the an ellipse is $A = \pi ab$.

- (d) The tangent at $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the directrix at Q . The focus of the parabola is S .

- (i) Find the equation of the tangent at P . **2**
(ii) Find the coordinates of Q . **1**
(iii) Show that PQ subtends a right angle at S . **4**

Question 13 (15 marks)**Marks**

- (a) When a submarine of mass m is travelling horizontally underwater at maximum power, its engine delivers a force F on this submarine. The water exerts a resistive force proportional to the square of the submarine's speed v .

(i) Explain why $\frac{dv}{dt} = \frac{1}{m}(F - kv^2)$, where k is a positive constant.

1

(ii) The submarine increases its speed from v_1 to v_2 . Show that the distance travelled during this period is

3

$$\frac{m}{2k} \log_e \frac{F - kv_1^2}{F - kv_2^2}$$

- (b) A particle of mass m is dropped from rest in a medium where the resistance to motion has magnitude $\frac{1}{10}mv^2$ Newtons. After t seconds, the particle has fallen x metres, and has the velocity $v \text{ ms}^{-1}$ and the acceleration $a \text{ ms}^{-2}$.

3

The particle hits the ground $\log_e(1 + \sqrt{2})$ seconds after it is dropped. Take $g = 10 \text{ ms}^{-2}$. Express v as a function of t .

- (c) A particle of mass 1 kg is moving horizontally in a straight line. It is initially at the origin and is moving with velocity $U \text{ ms}^{-1}$ ($U > 0$).

The particle is moving against a resistance $v^2 + v^3$, where v is the velocity.

After T seconds the particle is X metres from the origin and is moving with velocity $\frac{1}{2}U \text{ ms}^{-1}$.

(i) Show that $X = \log_e \left(\frac{2+U}{1+U} \right)$.

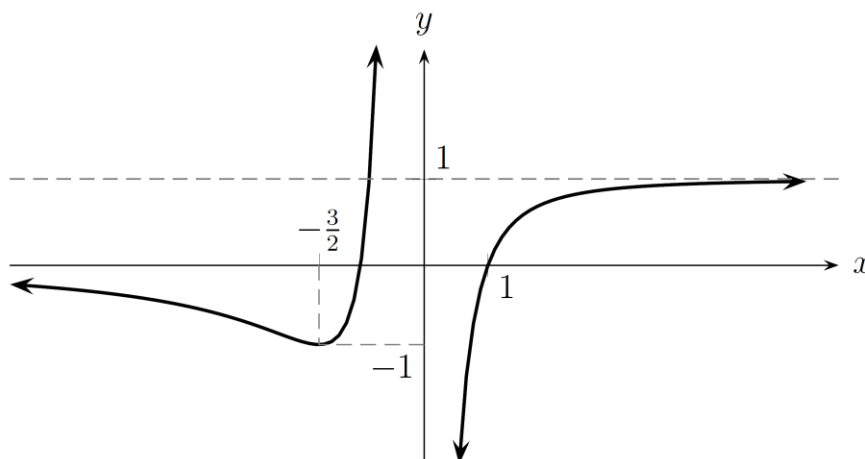
4

(ii) Show that $U(T + X) = 1$.

4

Question 14 (15 marks)**Marks**

- (a) The graph of $y = f(x)$ is shown below. The lines $y = 1$ and the x -axis are the asymptotes.



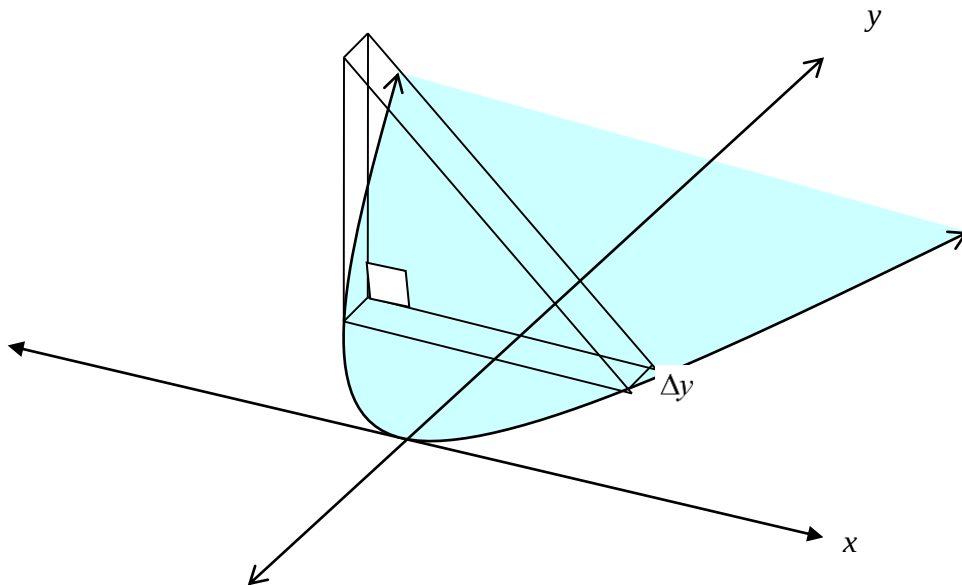
Draw a neat one-third page sketch of the following, showing relevant features:

- | | | |
|-------|----------------------|---|
| (i) | $y = f(-x)$ | 2 |
| (ii) | $y = f(x) $ | 2 |
| (iii) | $y = \frac{1}{f(x)}$ | 2 |
| (iv) | $y = (f(x))^2$ | 2 |

Question 14 continues overleaf...

Question 14 (continued...)

- (b) A solid is formed by rotating about the y -axis the region bounded by the curve $y = \sin x$ and the x -axis between $0 \leq x \leq \pi$. Find the volume of this solid using the method of cylindrical shells. 4
- (c) A solid shape is formed as shown below. Its base is in the xy plane and is in the shape of a parabola $y = x^2$. The vertical cross section is in the shape of a right angled isosceles triangle. 3



By using the method of slicing, calculate the volume of the solid between the values $y = 0$ and $y = 4$.

Examination continues overleaf...

Question 15 (15 marks)**Marks**

(a) Find $\int \frac{e^x}{\sqrt{e^{2x}-1}} dx$. 2

(b) Use the substitution $t = \tan \frac{x}{2}$ to evaluate 4

$$\int \frac{dx}{5 + 3 \sin x + 4 \cos x}$$

(c) (i) Find real numbers A , B , C and D such that: 2

$$\frac{4}{1-x^2} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2}$$

(ii) Hence evaluate in simplest form $\int \frac{4}{1-x^4} dx$ 2

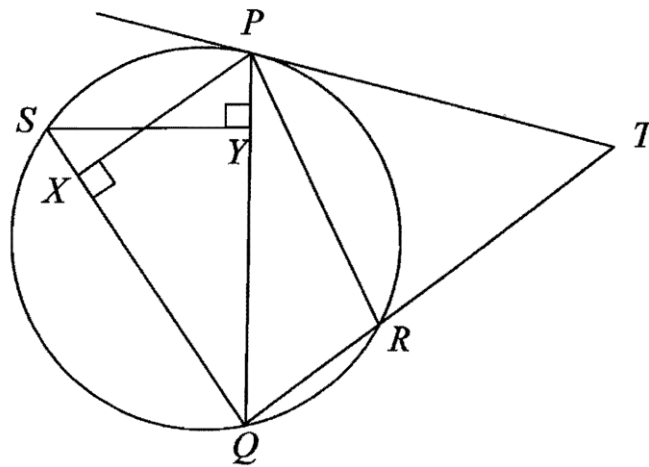
(d) (i) Let $I_n = \int_0^1 (1-x^r)^n dx$, where $r > 0$, for $n = 0, 1, 2, 3, \dots$ 3

Show that $I_n = \frac{nr}{nr+1} I_{n-1}$.

(ii) Hence or otherwise, find the value of $I_n = \int_0^1 (1-x^{\frac{3}{2}})^3 dx$. 2

Question 16 (15 marks)**Marks**

- (a) (i) Show that $\sin x + \sin 3x = 2 \sin 2x \cos 2x$ 1
 (ii) Hence or otherwise solve $\sin x + \sin 2x + \sin 3x = 0$ for $0 \leq x \leq 2\pi$. 2
- (b) Find the equation of the tangent to the curve $x^2y + 2x - 2xy = 0$ at $(1, 2)$. 2
- (c) In the diagram below, TP is a tangent to the circle at P , and TQ is a secant cutting the circle at R



SQ is a chord of the circle such that PX and SY are perpendicular to SQ and PQ respectively.

- (i) Prove that $\angle TRP = \angle TPQ$. 3
- (ii) Explain why $SPYX$ is a cyclic quadrilateral. 1
- (iii) Prove that $\angle PYX = \angle PRQ$. 3
- (d) $\triangle ABC$ has sides of length a , b and c . If $a^2 + b^2 + c^2 = ab + bc + ca$ show that $\triangle ABC$ is an equilateral triangle. 3

End of paper

Q11

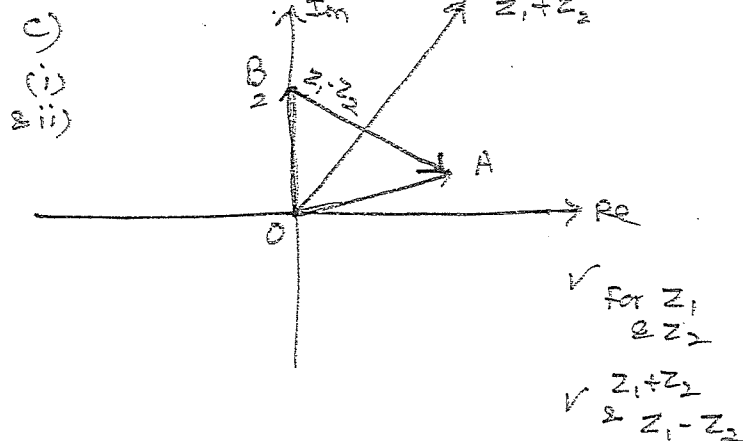
a) i) $(1-2i)^2$
 $= 1 - 4i + 4i^2$
 $= 1 - 4i - 4$
 $= -3 - 4i$ ✓

ii) $z^2 - 5z + (7+i) = 0$
 $z = \frac{5 \pm \sqrt{25 - 4(7+i)}}{2}$
 $= \frac{5 \pm \sqrt{25 - 28 - 4i}}{2}$
 $= \frac{5 \pm \sqrt{-3 - 4i}}{2}$
 $= \frac{5 \pm \sqrt{(1-2i)^2}}{2}$ ✓
 $= \frac{5 \pm (1-2i)}{2}$
 $= \frac{6-2i}{2}, \frac{4+2i}{2}$
 $= 3-i, 2+i$ ✓

b) $z = 2+2i$

i) $|z| = \sqrt{4+4}$
 $= 2\sqrt{2}$ ✓
 $\arg z = \frac{\pi}{4}$ ✓

ii) $z = 2\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$
 $z^3 = 16\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$ ✓
 $= 16\sqrt{2} \left(-\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right)$
 $= -16 + 16i$ ✓



iii) $\arg(z_1+z_2) = \frac{7\pi}{24}$

$\arg(z_1-z_2) = -\frac{5\pi}{24}$

d) i) $z^3 + 1 = 0$
 $z^3 = -1$

let $z = \text{cis } \theta$

$\text{cis } 3\theta = -1$

$\text{cis } 3\theta = \text{cis } (2n+1)\pi$

$3\theta = (2n+1)\pi$

$\theta = \frac{(2n+1)\pi}{3}$

as $-\pi < \theta < \pi$

$-\pi < \frac{(2n+1)\pi}{3} < \pi$

$-3\pi < (2n+1)\pi < 3\pi$

$-3 < (2n+1) < 3$

$-4 < 2n < 2$

$-2 < n < 1$

$\therefore n = -1, 0, 1$

$z_1 = \text{cis } \frac{(-2+1)\pi}{3} = \text{cis } \left(-\frac{\pi}{3}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$

$z_2 = \text{cis } \left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$

$z_3 = \text{cis } \left(\frac{3\pi}{3}\right) = \text{cis } \pi = -1$

$$) \alpha^3 + 1 = 0$$

$$(\alpha+1)(\alpha^2-\alpha+1)=0$$

as α is not the real root

$$\alpha+1 \neq 0$$

$$\therefore \alpha^2 - \alpha + 1 = 0$$

$$\alpha^2 = \alpha - 1$$

or Show it to be true with one of values of α .

$$\text{If } \alpha = \frac{1+\sqrt{3}i}{2}$$

$$\alpha^2 = \frac{-2+2\sqrt{3}i}{4}$$

$$= \frac{-1+\sqrt{3}i}{2}$$

$$= \frac{-2+1+\sqrt{3}i}{2}$$

$$\alpha^2 = -1 + \alpha$$

$$b+12=0$$

$$b = -12$$

Sub in (2)

$$4a - 12 + 32 = 0$$

$$4a + 20 = 0$$

$$a = -5$$

C)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

at P,

$$\frac{dy}{dx} = -\frac{b^2}{a^2} \frac{a \cos \theta}{b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

Equation of the tangent at P

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a \sin \theta y - ab \sin^2 \theta = -b \cos \theta x + ab \cos^2 \theta$$

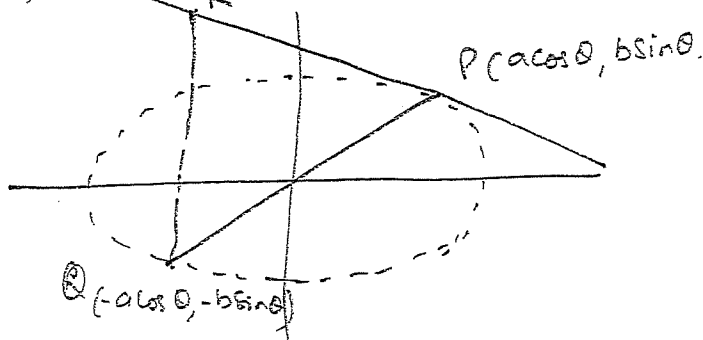
$$b \cos \theta x + a \sin \theta y = ab(\cos^2 \theta + \sin^2 \theta)$$

$$b \cos \theta x + a \sin \theta y = ab$$

Divide by ab

$$\frac{\cos \theta}{a} x + \frac{\sin \theta}{b} y = 1$$

ii)



$$i) (1-\alpha)^6 = (-(\alpha-1))^6$$

$$= (\alpha-1)^6$$

$$= \alpha^{12}$$

$$= (\alpha^3)^4$$

$$= (-1)^4 = 1 \quad \checkmark$$

Question 12

$$) 3x^3 - 5x^2 - 4x + 3 = 0$$

Hence required eq.

$$3(x+1)^3 - 5(x+1)^2 - 4(x+1) + 3 = 0$$

$$(x^3 + 3x^2 + 3x + 1) - 5(x^2 + 2x + 1) - 4x - 4 + 3 = 0$$

$$x^3 + 4x^2 + 9x + 3 - 5x^2 - 10x - 5 - 4x - 4 + 3 = 0$$

$$x^3 + 4x^2 - 5x - 3 = 0 \quad \checkmark$$

) Double root at 2

$$\Rightarrow P(2) = 0$$

$$P'(2) = 0$$

$$P(x) = x^4 + ax^2 + bx + 28$$

$$P'(x) = 4x^3 + 2ax + b$$

$$P(2) = 2^4 + a(2)^2 + b(2) + 28 = 0$$

$$= 16 + 4a + 2b + 28 = 0$$

$$4a + 2b + 44 = 0 \quad (1)$$

$$P'(2) = 4(2)^3 + 2a(2) + b = 0$$

$$= 32 + 4a + b = 0 \quad (2)$$

$$Q: (-a \cos \theta, -b \sin \theta)$$

$$\text{For R: } x: (-a \cos \theta)$$

Sub x-coordinate of R in equation of PQ

$$\frac{\cos \theta}{a} (-a \cos \theta) + \frac{\sin \theta}{b} y = 1$$

$$-\cos^2 \theta + \frac{\sin \theta}{b} y = 1$$

$$y = \frac{b(1 + \cos^2 \theta)}{\sin \theta}$$

Length RQ

$$= b \frac{(1 + \cos^2 \theta)}{\sin \theta} + b \sin \theta$$

$$= \frac{b + b \cos^2 \theta + b \sin^2 \theta}{\sin \theta}$$

$$= \frac{2b}{\sin \theta}$$

$$\left. \begin{array}{l} \text{Perpendicular distance of RQ from P} \\ = 2a \cos \theta \end{array} \right\}$$

Area of ΔPQR

$$= \frac{1}{2} \times \frac{2b}{\sin \theta} \times 2a \cos \theta$$

$$= 2ab \cot \theta$$

$$= \frac{2ab}{\tan \theta} = \frac{2ab}{|\tan \theta|} \quad \text{as area} > 0$$

Ratio of ar ΔPQR to ellipse

$$= \frac{2ab}{|\tan \theta|} \div \pi ab$$

$$= 2 : \pi |\tan \theta|$$

$$a) \quad i) \quad \frac{x}{a^2} - \frac{y}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2x}{a^2} \times \frac{b^2}{2y}$$

$$= \frac{b^2 x}{a^2 y}$$

$$\text{at P,} \quad \frac{dy}{dx} = \frac{b^2 \cdot a \sec \theta}{a^2 \cdot b \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

Eq. of the tangent at P

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$a \tan \theta y - ab \tan^2 \theta = b \sec \theta x - ab \sec^2 \theta$$

$$b \sec \theta x - a \tan \theta y = ab(\sec^2 \theta - \tan^2 \theta)$$

$$b \sec \theta x - a \tan \theta y = ab$$

Divide by ab

$$\frac{\sec \theta}{a} x - \frac{\tan \theta}{b} y = 1$$

ii) Eq. of directrix, $x = \frac{a}{e}$

at Q, eq. of PQ becomes

$$\frac{\sec \theta}{a} \cdot \frac{a}{e} - \frac{\tan \theta}{b} y = 1$$

$$- \frac{\tan \theta}{b} y = 1 - \frac{\sec \theta}{e}$$

$$y = \left(\frac{\sec \theta}{e} - 1 \right) \frac{b}{\tan \theta}$$

$$= \frac{b \sec \theta}{e \tan \theta} - \frac{b}{\tan \theta}$$

$\therefore$ for Q: $x = \frac{a}{e}$

$$y = \frac{b \sec \theta}{e \tan \theta} - \frac{b}{\tan \theta}$$

iii) P: ($a \sec \theta$, $b \tan \theta$)

Q: ($\frac{a}{e}$, $\frac{b \sec \theta}{e \tan \theta} - \frac{b}{\tan \theta}$)

S: (ae , 0)

Gradient of PS

$$m_{PS} = \frac{b \tan \theta}{a \sec \theta - ae}$$

$$= \frac{b \tan \theta}{a(\sec \theta - e)}$$

Gradient of QS

$$m_{QS} = \frac{\frac{b \sec \theta}{e \tan \theta} - \frac{b}{\tan \theta}}{\frac{a}{e} - ae}$$

$$= \frac{\frac{b \sec \theta - be}{e \tan \theta}}{\frac{a - ae^2}{e}}$$

$$= \frac{b(\sec \theta - e)}{e \tan \theta} \times \frac{e}{a(1 - e^2)}$$

$$= \frac{b(\sec \theta - e)}{a(1 - e^2) \tan \theta}$$

$m_{PS} \times m_{QS}$

$$= \frac{b \tan \theta}{a(\sec \theta - e)} \times \frac{b(\sec \theta - e)}{a(1 - e^2) \tan \theta}$$

$$= \frac{b^2}{a^2(1 - e^2)}$$

$$= \frac{b^2}{-a^2(e^2 - 1)}$$

$$= -\frac{b^2}{b^2}$$

$$= -1$$

Question 13

a) $ma = F - kv^2$

i) (Net force = Forward force - Resistance) ✓

$$a = \frac{1}{m} (F - kv^2)$$

$$\frac{dv}{dt} = \frac{1}{m} (F - kv^2)$$

ii) $a = \frac{1}{m} (F - kv^2)$

$$v \cdot \frac{dv}{dx} = \frac{1}{m} (F - kv^2)$$

$$\frac{dv}{dx} = \frac{F - kv^2}{mv}$$

$$\frac{dx}{dv} = \frac{mv}{F - kv^2}$$

$$x = m \int \frac{v}{F - kv^2} dv$$
 ✓

$$= -\frac{m}{2k} \int_{v_1}^{v_2} \frac{2kv}{F - kv^2} dv$$

$$x = -\frac{m}{2k} \left[\ln(F - kv^2) \right]_{v_1}^{v_2}$$
 ✓

$$= -\frac{m}{2k} \left[\ln(F - kv_2^2) - \ln(F - kv_1^2) \right]$$

$$= \frac{m}{2k} \left[\ln(F - kv_1^2) - \ln(F - kv_2^2) \right]$$
 ✓

$$= \frac{m}{2k} \left[\ln \frac{F - kv_1^2}{F - kv_2^2} \right]$$

b) $v \downarrow$ $mg \uparrow R$

$$ma = mg - R$$

$$ma = mg - \frac{1}{10}mv^2$$

$$a = g - \frac{1}{10}v^2$$

$$a = 10 - \frac{1}{10}v^2$$

$$a = \frac{100 - v^2}{10}$$

$$\frac{dv}{dt} = \frac{100 - v^2}{10}$$

$$t = \int \frac{10}{100 - v^2} dv$$

$$t = 10 \int \frac{1}{100 - v^2} dv$$

$$\text{for } \frac{1}{100 - v^2} = \frac{A}{(10 - v)(10 + v)}$$

$$= \frac{A}{10 - v} + \frac{B}{10 + v}$$

$$A(10 + v) + B(10 - v) = 1$$

when $v = 10$

$$20A = 1$$

$$A = \frac{1}{20}$$

if $v = -10$

$$20B = 1$$

$$B = \frac{1}{20}$$

$$t = 10 \left[\frac{1}{20(10 - v)} + \frac{1}{20(10 + v)} \right] dv$$

$$= \frac{1}{2} \int \left(\frac{1}{10 - v} + \frac{1}{10 + v} \right) dv$$

$$= \frac{1}{2} \left[-\ln(10 - v) + \ln(10 + v) \right] + C$$

$$= \frac{1}{2} \left[\ln \frac{10 + v}{10 - v} \right] + C$$

when $t = 0$, $v = 0 \Rightarrow C = 0$

$$\therefore t = \frac{1}{2} \left[\ln \frac{10 + v}{10 - v} \right]$$

$$\therefore e^{2t} = \frac{10 + v}{10 - v}$$

$$(10 - v)e^{2t} = 10 + v$$

$$10e^{2t} - ve^{2t} = 10 + v$$

$$10(e^{2t} - 1) = v + ve^{2t}$$

$$v(1 + e^{2t}) = 10(e^{2t} - 1)$$

$$v = \frac{10(e^{2t} - 1)}{e^{2t} + 1}$$

c)

$$ma = -m(v^2 + v^3)$$

$$a = -(v^2 + v^3)$$

$$v \cdot \frac{dv}{dx} = -(v^2 + v^3)$$

$$\frac{dv}{dx} = -(v + v^2)$$

$$x = \int \frac{-1}{v + v^2} dv$$

$$= - \int \frac{1}{v(1 + v)} dv$$

$$\text{For } \frac{1}{v(1 + v)} = \frac{A}{v} + \frac{B}{1 + v}$$

$$A(1 + v) + Bv = 1$$

when $v = 0$

$$A = 1$$

when $v = 1$

$$2A + B = 1$$

$$B = -1$$

$$\therefore x = - \int \frac{1}{v} - \frac{1}{1 + v} dv$$

$$= \int \frac{1}{1 + v} - \frac{1}{v} dv$$

$$= \left[\ln(1 + v) - \ln v \right] + C$$

$$x = \ln \left[\frac{1 + v}{v} \right] + C$$

when $x = 0$, $v = V$

$$0 = \ln \left[\frac{1 + V}{V} \right] + C$$

$$\therefore x = \ln \left[\frac{1 + v}{v} \right] - \ln \left[\frac{1 + V}{V} \right]$$

$$= \ln \left[\frac{(1 + v)(V)}{v(1 + V)} \right]$$

(1)

When $x = X$, $v = \frac{1}{2} U$

∴ From (1)

$$X = \ln \left[\frac{1 + \frac{1}{2}U}{\frac{1}{2}U} \cdot \frac{U}{(1+U)} \right]$$

$$= \ln \left[\frac{2+U}{2} \cdot \frac{2}{U} \cdot \frac{U}{(1+U)} \right]$$

$$X = \ln \left[\frac{2+U}{1+U} \right]$$

ii) $a = -(v^2 + v^3)$

$$\frac{dv}{dt} = -(v^2 + v^3)$$

$$t = \int \frac{-1}{v^2 + v^3} dv$$

$$= - \int \frac{1}{v^2(1+v)} dv$$

For $\frac{1}{v^2(1+v)} = \frac{A}{v} + \frac{B}{v^2} + \frac{C}{1+v}$

$$(Av+B)(1+v) + Cv^2 = 1$$

When $v = 0$

$$\boxed{B = 1}$$

When $v = -1$

$$\boxed{C = 1}$$

When $v = 1$

$$(A+B)2 + C = 1$$

$$2A + 2 + 1 = 1$$

$$\boxed{A = -1}$$

$$t = - \int \frac{-v+1}{v^2} + \frac{1}{1+v} dv$$

$$= - \int \frac{1-v}{v^2} + \frac{1}{1+v} dv$$

$$= - \int \frac{1}{v^2} - \frac{1}{v} + \frac{1}{1+v} dv$$

$$= \int \frac{1}{v} - \frac{1}{v^2} - \frac{1}{1+v} dv$$

$$t = \ln v + \frac{1}{v} - \ln(1+v) + C$$

$$t = \frac{1}{v} + \ln \frac{v}{1+v} + C$$

When $t = 0$, $v = U$

$$0 = \frac{1}{U} + \ln \frac{U}{1+U} + C$$

$$\therefore t = \frac{1}{v} + \ln \frac{v}{1+v} - \frac{1}{U} - \ln \frac{U}{1+U}$$

$$= \frac{1}{v} - \frac{1}{U} + \ln \frac{v(1+U)}{(1+v)U}$$

When $t = T$, $v = \frac{1}{2} U$

$$T = \frac{1}{\frac{U}{2}} - \frac{1}{U} + \ln \frac{\frac{U}{2}(1+U)}{(1+\frac{U}{2})U}$$

$$= \frac{2}{U} - \frac{1}{U} + \ln \frac{U(1+U)}{2(2+U)U}$$

$$T = \frac{1}{U} + \ln \frac{1+U}{2+U}$$

$$= \frac{1}{U} + \ln \left(\frac{2+U}{1+U} \right)^{-1}$$

$$T = \frac{1}{U} - \ln \left(\frac{2+U}{1+U} \right)$$

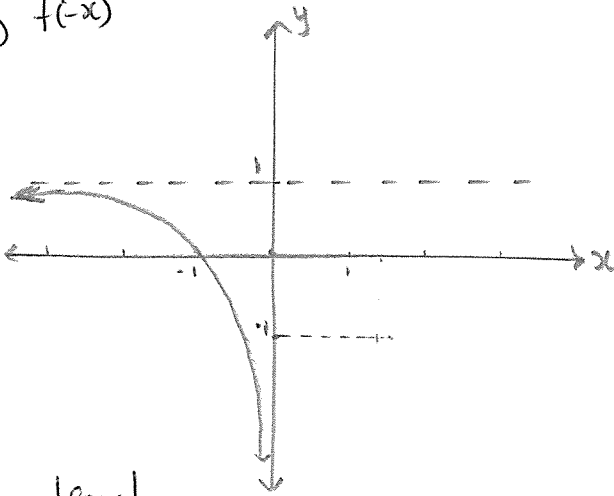
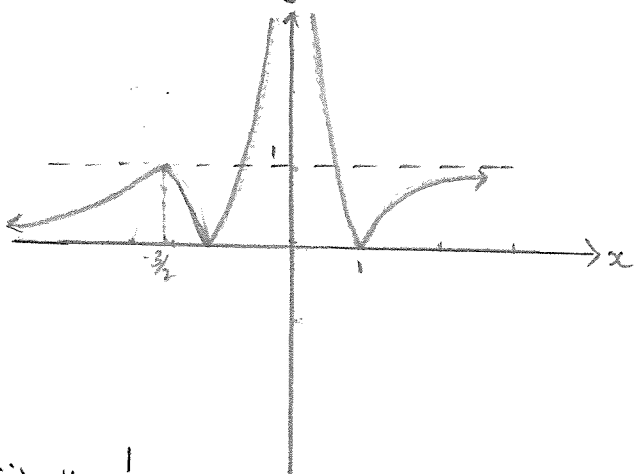
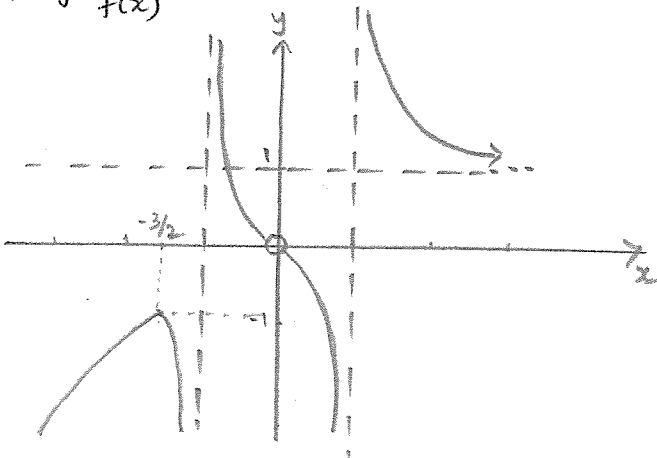
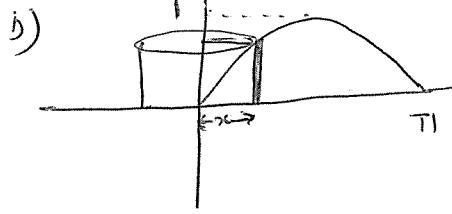
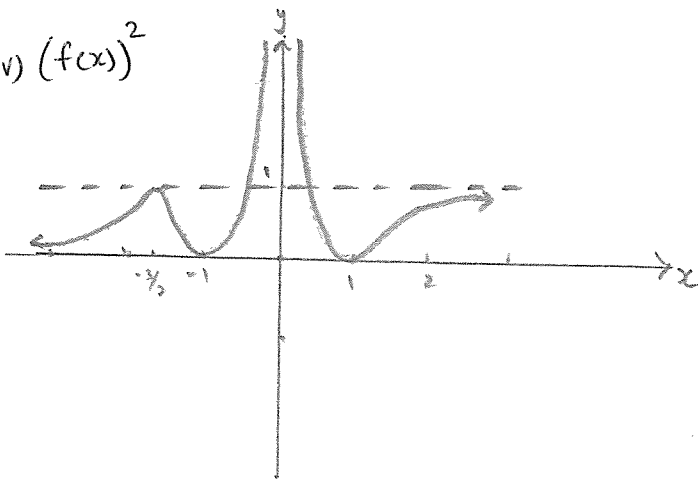
From part (i)

$$T = \frac{1}{U} - X$$

$$T + X = \frac{1}{U}$$

$$\therefore U(T+X) = 1$$

Q14

a) i) $f(-x)$ ii) $y = |f(x)|$ iii) $y = \frac{1}{f(x)}$ iv) $(f(x))^2$ 

For the shell

radius $= r = x$

height $= h = y = \sin x$

thickness $= \delta x$

$$\delta V = 2\pi x \cdot h \cdot \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi} \delta V$$

$$= 2\pi \int_0^{\pi} x \sin x \, dx$$

For $\int_0^{\pi} x \sin x \, dx$

let $u = x$, $v = -\cos x$

$u' = 1$, $v' = \sin x$

$$\therefore \int_0^{\pi} x \sin x \, dx = \left[-x \cos x \right]_0^{\pi} + \int_0^{\pi} \cos x \, dx$$

$$= \left[-\pi \cos \pi \right] + \left[\sin x \right]_0^{\pi}$$

$$= \pi$$

$$\therefore V = 2\pi \cdot \pi$$

$$= 2\pi^2 \text{ units}^3$$

c) For triangular cross-section
base $= 2x = \text{height}$ $\therefore$ area of the cross-section

$$A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 2x \times 2x$$

$$= 2x^2$$

$$\delta V = \text{Area of cross-section} \times \text{thickness}$$

$$\delta V = 2x^2 \cdot \delta y$$

$$V = \int_0^4 2x^2 \, dy$$

$$= \int_0^4 2y \, dy$$

$$= \left[\frac{2y^2}{2} \right]_0^4 = 16 \text{ units}^3$$

Question 15

i) $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$

let $e^x = u$

$e^x = \frac{du}{dx}$

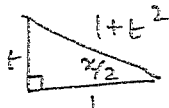
i. $\int \frac{1}{\sqrt{1-u^2}} du$ ✓

$= \sin^{-1} u + C$

$= \sin^{-1}(e^x) + C$ ✓

b) $t = \tan \frac{x}{2}$

$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$



$\frac{dt}{dx} = \frac{1+t^2}{2}$

$\therefore dx = \frac{2}{1+t^2} dt$ ✓

$\Rightarrow \int \frac{dx}{5+3\sin x + 4\cos x}$

$= \int \frac{1}{5+3\left(\frac{2t}{1+t^2}\right)+4\left(\frac{1-t^2}{1+t^2}\right)} \cdot \left(\frac{2}{1+t^2}\right) dt$ ✓

$= \int \frac{\cancel{1+t^2}}{5(1+t^2)+6t+4(1-t^2)} \cdot \frac{2}{\cancel{1+t^2}} dt$

$= \int \frac{2}{5+5t^2+6t+4-4t^2} dt$

$= \int \frac{2}{t^2+6t+9} dt$ ✓

$= \int \frac{2}{(t+3)^2} dt$

$= 2 \int (t+3)^{-2} dt$

$= \frac{-2}{t+3} + C$

$= \frac{-2}{\tan \frac{x}{2} + 3} + C$ ✓

c) $\frac{4}{1-x^4} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2}$

(i) $\Rightarrow A(1+x)(1+x^2) + B(1-x)(1+x^2) + (Cx+D)(1-x^2) = 4$

If $x=0$

$A+B+D=4$ - (1)

If $x=1$

$4A=4$

$A=1$

If $x=-1$

$4B=4$

$B=1$

From (1) $D=2$

If $x=2$

$A(3)(5) + B(-1)(5) + (2C+D)(-3) = 4$

$15A - 5B - 6C - 6 = 4$

$15 - 5 - 6C - 6 = 4$

$C=0$

$\therefore \frac{4}{1-x^4} = \frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2}$

(ii) $\int \frac{4}{1-x^4} dx = \int \frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} dx$

$= -\ln(1-x) + \ln(1+x) + 2 \tan^{-1} x + C$ ✓

$= \ln \frac{1+x}{1-x} + 2 \tan^{-1} x + C$ ✓

✓ values of A, B, C, D
✓ (1 mark off for any incorrect value)

i) $I_n = \int_0^1 (1-x^r)^n dx$

let $u = (1-x^r)^n$

$$u' = n(1-x^r)^{n-1} \cdot (-rx^{r-1})$$

$$= -nr x^{r-1} (1-x^r)^{n-1}$$

and $v' = 1$

$\therefore v = x$

as $\int uv' = uv - \int vu'$

$$I_n = \left[(1-x^r)^n \cdot x \right]_0^1 + \int_0^1 nr x^{r-1} (1-x^r)^{n-1} \cdot x dx$$

$$= nr \int_0^1 x^r (1-x^r)^{n-1} dx$$

$$= nr \int_0^1 (1 - 1 + x^r) (1-x^r)^{n-1} dx$$

$$= nr \int_0^1 (1 - (1-x^r)) (1-x^r)^{n-1} dx$$

$$= nr \int_0^1 (1-x^r)^{n-1} - (1-x^r)^n dx$$

$$= nr \int_0^1 (1-x^r)^{n-1} dx - nr \int_0^1 (1-x^r)^n dx$$

$$I_n = nr I_{n-1} - nr I_n$$

$$I_n (1+nr) = nr I_{n-1}$$

$$I_n = \frac{nr}{1+nr} I_{n-1}$$

ii) $I_n = \int_0^1 (1-x^{\frac{3}{2}})^n dx$
 $r = \frac{3}{2}, n \geq 3$

$$I_n = I_3 = \frac{3(\frac{3}{2})}{1+\frac{9}{2}} \cdot I_2$$

$$= \frac{9}{2} \cdot \frac{2}{11} I_2$$

$$= \frac{9}{11} I_2$$

$$= \frac{9}{11} \left[\frac{2 \cdot \frac{3}{2}}{1+2 \cdot \frac{3}{2}} \cdot I_1 \right]$$

$$= \frac{9}{11} \left[\frac{3}{4} I_1 \right]$$

$$= \frac{9}{11} \left(\frac{3}{4} \right) \left(\frac{\frac{3}{2}}{1+\frac{3}{2}} I_0 \right)$$

$$= \frac{9}{11} \times \frac{3}{4} \times \frac{3}{5} I_0$$

for $I_0 = \int_0^1 (1-x^{\frac{3}{2}})^0 dx$

$$= \int_0^1 1 dx$$

$$= [x]_0^1$$

$$= 1$$

$$\therefore I_n = I_3 = \frac{9}{11} \times \frac{3}{4} \times \frac{3}{5}$$

$$= \frac{81}{220}$$

Question 16

i) $\sin x + \sin 3x$

$$= 2 \sin\left(\frac{x+3x}{2}\right) \cos\left(\frac{x-3x}{2}\right)$$

$$= 2 \sin \frac{4x}{2} \cos\left(-\frac{2x}{2}\right)$$

$$= 2 \sin 2x \cos(-x)$$

$$= 2 \sin 2x \cos x$$

ii) $\sin x + \sin 2x + \sin 3x = 0$

$$\sin 2x + 2 \sin 2x \cos x = 0$$

$$\sin 2x (1 + 2 \cos x) = 0$$

$\therefore \sin 2x = 0$

$$1 + 2 \cos x = 0$$

$$2x = 0, \pi, 2\pi$$

$$\cos x = -\frac{1}{2}$$

$$x = 0, \frac{\pi}{2}, \pi$$

$$x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$= \frac{2\pi}{3}, \frac{4\pi}{3}$$

✓

✓

b) $x^2 y + 2x - 2xy = 0$

$$y(x^2 - 2x) = -2x$$

$$y = \frac{-2x}{x^2 - 2x}$$

$$= \frac{-2}{x-2}$$

$$\frac{dy}{dx} = \frac{(x-2)(0) - (-2)}{(x-2)^2}$$

$$= \frac{2}{(x-2)^2}$$

at (1,2)

$$\frac{dy}{dx} = \frac{2}{1} = 2$$

q. of tangent

$$y-2 = 2(x-1)$$

$$2x-4=0$$

✓

c) i) $\angle PSQ + \angle PRQ = 180^\circ$

of a cyclic quadrilateral

But

$$\angle PRQ + \angle PRT = 180^\circ$$

(adjacent supplementary angles)

$$\therefore \angle PSQ = \angle PRT$$

However

$$\angle PSQ = \angle TPQ$$

(angles in alt. segments)

$$\therefore \angle TPQ = \angle TRP$$

ii)

$$\angle SXP = \angle SYP = 90^\circ$$

(converse of angles in the same segment)

iii)

$$\angle PYX + \angle PSX = 180^\circ$$

(opp. angles of a cyclic quadrilateral)

$$\angle PYX = 180 - \angle PSX$$

$$= 180^\circ -$$

But $\angle PSX + \angle PRQ = 180^\circ$

(opp. angles of a cyclic quad. PSQR)

$$\Rightarrow \angle PSX = 180^\circ - \angle PRQ$$

$\therefore$ from ①

$$\angle PYX = 180 - (180^\circ - \angle PRQ)$$

$$\angle PYX = \angle PRQ$$

$$(a-b)^2 + (b-c)^2 + (a-c)^2$$

$$= a^2 - 2ab + b^2 + b^2 - 2bc + c^2 + a^2 - 2ac + c^2$$

$$= 2(a^2 + b^2 + c^2) - 2ab - 2bc - 2ac$$

$$= 2(a^2 + b^2 + c^2 - ab - bc - ac)$$

$$= 0 \quad \text{because} \quad a^2 + b^2 + c^2 = ab + bc + ac \quad (\text{given})$$

Possible only if

$$a-b=0 \quad b-c=0 \quad c-a=0$$

$$\Rightarrow a=b, \quad b=c, \quad c=a$$

$\therefore \Delta ABC$ is an equilateral triangle.

MCCQ

Q 1	A	6	B
2	C	7	B
3	B	8	B
4	B	9	C
5	D	10	B